

Commuting Shed Similarity via LODES Origin-Destination Data

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Abstract

Where people commute defines their economic community more precisely than where they work or where they live considered in isolation. Two residential census tracts whose workers flow to the same employment centers share a functional economic area: they patronize the same commercial corridors, depend on the same transportation infrastructure, and have a shared stake in the policy decisions that govern their common destination. This paper presents M.4, the fourth paper in the M-track community character series, which operationalizes commuting community membership through the Census Bureau’s LODES Origin-Destination (OD) file.

For each census tract we compute a commuting destination distribution $D(h) = \{w : \text{count}(h \rightarrow w)\}$: a sparse vector recording how many workers residing in tract h are employed in each destination tract w . The community similarity between two home tracts is the weighted Jaccard similarity of their destination distributions: the overlap of commute flows divided by the union of commute flows. Tracts whose workers commute to the same employment centers receive high similarity and heavy edge weights; tracts serving different economic spheres receive low similarity and light edges that invite district cuts at functional economic area boundaries.

Experimental design targets Wisconsin ($k = 8$) and North Carolina ($k = 14$) with primary metrics of suburban clustering rates and polycentric separation. The Wisconsin prediction—Milwaukee suburb tracts sharing a primary destination co-assign at $\geq 70\%$ rate—and the North Carolina prediction—Raleigh metro and Charlotte metro tracts achieve cross-metro co-assignment below 5% under commuting-shed weights—are both deferred to Phase 2 pending LODES OD data download.[†] The LODES OD data is annual, free, publicly available, and contains no racial or partisan content.

Keywords

Commuting shed, LODES, Origin-destination, Functional economic area, Edge weights, METIS, Weighted Jaccard similarity, Redistricting

1 Introduction

The citizens of Waukesha County, Wisconsin testified before the Wisconsin Redistricting Commission in 2021 that their communities belonged in the same legislative district as the other Milwaukee suburban counties to the north and east. Their argument was not about geography alone—Waukesha is not adjacent to all the communities they claimed to share a district with. Their argument was about daily life: tens of thousands of Waukesha residents drive the same freeways, park in the same lots, and work in the same office parks in Milwaukee’s western suburbs. They were, they testified, part of the same economic community. The Commission heard the testimony, found it compelling, but had no quantitative tool to measure the strength of the commuting community bond.

This paper provides that measurement.

The commuting shed concept has a long history in regional science. Tolbert and Sizer [1996] defined functional economic areas on the basis of inter-county commuting flows, arguing that the daily commuting sphere—not the municipal or county boundary—is the appropriate unit of economic community. This work has been applied to labor market delineation, retail trade area analysis, and economic development planning. But it has not been applied to redistricting, where the relevant question is not “what is the optimal functional economic area size?” but “do these two adjacent tracts belong to the same economic community?”

The M-track community character framework, introduced in M.0 [Della-Libera, 2025], operationalizes diverse community signals as edge weights in the census-tract adjacency graph. M.4 is the commuting shed implementation paper. It uses the Census Bureau’s LODES Origin-Destination (OD) file—which records home-block to work-block commute flows for all UI-covered employees—to compute a per-tract commuting destination distribution and measure the similarity between adjacent tracts’ distributions using the weighted Jaccard coefficient.

Commuting shed similarity is the right M-track signal for polycentric metropolitan areas for several reasons:

1. **Functional community, not just residential proximity.** Two adjacent census tracts may be geographically contiguous but serve completely different employment centers. Tract A’s workers commute north to a technology corridor; tract B’s workers

commute south to a port logistics hub. Geographic proximity implies no community bond; commuting destination overlap captures the functional difference.

2. **Revealed preference.** Unlike survey-based community identification or administrative zone membership, commute flows are behavioral data. They reflect what workers actually do, not what planners designate or what residents aspire to. The commuting shed is the economic community as expressed through daily behavior.
3. **Complementary to M.1.** M.1 (economic character, [Della-Libera, 2026a]) weights tracts by their employment profile: where jobs are located. M.4 weights tracts by their residential commuting profile: where workers from each tract actually go. M.1 identifies employment centers; M.4 identifies their residential catchment areas. Together they provide a complete picture of the economic community.
4. **Partisan neutrality.** LODES OD data contains no electoral, racial, or demographic information. The commute flow records are administrative data derived from state UI wage records matched to employer geocodes.

Distinction from M.1 economic character. A high-employment tract (M.1 jobs per resident > 5) represents an employment center. M.4 identifies which residential tracts send workers to that center. Two residential tracts that both send workers primarily to the same tech park have a commuting-shed community bond; two residential tracts that send workers to different employment centers do not—even if they are adjacent. The M.4 signal fires where M.1 is silent: in residential areas with low JPR values that cannot be distinguished by economic character but can be distinguished by the direction of their daily commute.

Paper organization. Section 2 describes the LODES OD data structure and the functional economic area concept. Section 3 defines the commuting destination distribution, weighted Jaccard similarity, sparsity handling, and edge weight formula. Section 4 describes the experimental design and defers quantitative results to Phase 2. Section 5 provides a legal usage guide including airport communities, port communities, and economic commuting shed recognition. Section 6 concludes.

2 Background and Data

2.1 LODES Origin-Destination Data

The Longitudinal Employer-Household Dynamics (LEHD) program at the U.S. Census Bureau compiles the LODES (LEHD Origin-Destination Employment Statistics) dataset from state Unemployment Insurance wage records matched to employer and residential address geocodes [Abowd et al., 2009].

The LODES dataset has three file types:

- **WAC** (Workplace Area Characteristics): jobs by census block of work, used in M.1 [Della-Libera, 2026a].
- **RAC** (Residence Area Characteristics): workers by census block of residence.
- **OD** (Origin-Destination): job counts by home block and work block pair.

M.4 uses the OD file, which is the most granular LODES product.

OD file structure. The OD main file (job type JT00, all jobs) has the following primary columns:

- **h_geocode** (15-character block GEOID of home location)
- **w_geocode** (15-character block GEOID of work location)
- **S000** (total number of jobs with this home-work pair)

Additional columns (**SA01–SA03**) break the job count by age group; these are not used in M.4. Each row records how many workers live in block **h_geocode** and work in block **w_geocode**. The 2020 vintage is used. File naming convention:

`{state}_od_main_JT00_2020.csv.gz`

available at <https://lehd.ces.census.gov/data/lodes/LODES8/{state}/od/>.

Block-to-tract aggregation. Both **h_geocode** and **w_geocode** are 15-character block GEOIDs. The first 11 characters identify the census tract (state 2 + county 3 + tract 6). Block-to-tract aggregation proceeds in two steps:

1. Truncate **h_geocode** and **w_geocode** to 11 characters to obtain home tract h and work tract w .

2. Sum S000 for all block pairs within the same (h, w) tract pair.

The result is a sparse matrix F where $F[h, w]$ is the total number of workers residing in home tract h and employed in work tract w .

File size and sparsity. The OD file for a large state can be substantial. The 2020 NC OD file has approximately 2.1 million rows before tract aggregation; after aggregation to the 2,655 NC census tracts, the resulting matrix has at most $2655^2 \approx 7\text{M}$ cells but is highly sparse. The average NC home tract has commuting flows to approximately 15–30 distinct work tracts; most home-work tract pairs have zero flow. This sparsity is exploited in the weighted Jaccard computation (Section 3).

Cross-state flows. LODES OD files are produced per state and record only intra-state commuting flows. Cross-state commuters (e.g., New Jersey residents commuting to New York City) are partially captured in both states’ files but with geographic coverage gaps. For single-state redistricting runs, cross-state flow omission means that border tracts may have artificially low commuting destination overlap with tracts that would be captured if inter-state flows were included. This is an acceptable limitation for the purpose of the M.4 edge weight: the within-state commuting pattern still provides a meaningful community signal, and border-state anomalies affect a small fraction of tracts.

2.2 Functional Economic Areas and Commuting Sheds

Tolbert-Sizer functional economic areas. Tolbert and Sizer [1996] defined functional economic areas (FEAs) for the entire United States by identifying county-level “nodal clusters” where commuting flows indicated economic integration. Their 1996 analysis produced 741 FEAs covering all U.S. counties. The methodology has been updated periodically and provides the intellectual foundation for Metropolitan Statistical Area delineation by the Office of Management and Budget.

FEAs are defined at the county level; M.4 extends this concept to the census tract level. Two adjacent census tracts that share a commuting destination cluster belong to the same functional economic community at the sub-county scale. This captures a finer-grained community structure than FEA analysis: within a single FEA, there may be multiple distinct sub-communities with different primary employment destinations.

Research Triangle as a test case. The Research Triangle Park (RTP) area in North Carolina provides a well-studied example of commuting shed community structure. Workers residing in Raleigh, Durham, Chapel Hill, Cary, and Research Triangle Park itself all

commute to the same concentration of pharmaceutical, biotechnology, and information technology employers located in the RTP complex and the I-40/US-1 corridor. This creates a natural commuting shed that spans four counties (Wake, Durham, Orange, Chatham) but excludes the Charlotte metro tracts 100 miles to the southwest, which commute to a completely different employment cluster (finance in Uptown Charlotte, distribution in the I-85/I-77 corridor).

The M.4 algorithm is expected to assign high weighted Jaccard similarity to Wake-Durham-Orange tract pairs that share RTP as a primary destination and low similarity to pairs that span the RTP/Charlotte employment divide. This prediction is the basis for the NC-14 polycentric separation test in Section 4.

Milwaukee suburban commuting sheds. Waukesha County (west of Milwaukee) and the northern Milwaukee suburbs (Ozaukee, Washington counties) both exhibit high commuting flows to downtown Milwaukee and the western suburban employment corridor (Brookfield, Wauwatosa, Menomonee Falls). Despite not being geographically contiguous with each other, tracts in these three counties share destination overlap through their common primary destinations. The M.4 algorithm is expected to assign similarity scores above 0.5 to pairs of tracts from these counties that share Milwaukee-area primary destinations.

3 Algorithm

3.1 Commuting Destination Distribution

For each home census tract h , the commuting destination distribution is a sparse vector:

$$D(h) = \{w : F[h, w] > 0\} \quad \text{with values } D(h)[w] = F[h, w] \quad (1)$$

where $F[h, w]$ is the number of workers residing in tract h and employed in tract w , aggregated from the LODES OD block-level records as described in Section 2.

The distribution is stored in unnormalized count form. Normalization to a probability distribution (dividing by $\sum_w D(h)[w]$) is performed at similarity computation time, not at storage time, to avoid floating-point precision issues in the sparse representation.

3.2 Weighted Jaccard Similarity

The community similarity between two home tracts h_1 and h_2 is the weighted Jaccard similarity of their commuting destination distributions:

$$\text{sim}(h_1, h_2) = \frac{\sum_w \min(D(h_1)[w], D(h_2)[w])}{\sum_w \max(D(h_1)[w], D(h_2)[w])} \quad (2)$$

where the sums run over all work tracts w that appear in either $D(h_1)$ or $D(h_2)$. The min and max functions operate element-wise: for w present in $D(h_1)$ but not $D(h_2)$, $\min(\cdot) = 0$ and $\max(\cdot) = D(h_1)[w]$.

Properties of weighted Jaccard similarity.

- **Bounded.** $\text{sim}(h_1, h_2) \in [0, 1]$ for all pairs.
- **Symmetric.** $\text{sim}(h_1, h_2) = \text{sim}(h_2, h_1)$ because min and max are symmetric.
- **Self-similarity.** $\text{sim}(h, h) = 1.0$ for any tract with at least one commuting flow.
- **Disjoint zero.** $\text{sim}(h_1, h_2) = 0.0$ when $D(h_1)$ and $D(h_2)$ have no destination tract in common ($\sum \min = 0$).

Interpretation. The weighted Jaccard similarity measures the overlap of commuting flows rather than binary destination co-membership. If h_1 sends 100 workers to downtown Milwaukee and 20 to Brookfield, and h_2 sends 80 workers to downtown Milwaukee and 60 to Brookfield:

$$\sum_w \min = \min(100, 80) + \min(20, 60) = 80 + 20 = 100 \quad (3)$$

$$\sum_w \max = \max(100, 80) + \max(20, 60) = 100 + 60 = 160 \quad (4)$$

$$\text{sim}(h_1, h_2) = 100/160 = 0.625 \quad (5)$$

This is a meaningful community bond: both tracts primarily commute to the same Milwaukee employment areas, even though the proportions differ.

If the same h_1 is compared to a tract h_3 that sends workers to Port of Milwaukee logistics hubs rather than downtown Milwaukee: $\text{sim}(h_1, h_3) \approx 0$ because the destination sets are disjoint. The algorithm correctly identifies the community boundary.

3.3 Computational Complexity

For a home tract h , the commuting distribution $D(h)$ has at most W non-zero entries, where W is the number of distinct work tracts receiving commute flows from h . Empirically, $W \approx 15\text{--}30$ per home tract based on LODES OD density analysis for mid-sized states.

The weighted Jaccard for a pair (h_1, h_2) requires iterating over the union of their destination sets, which has at most $W_1 + W_2$ elements. For all n adjacent tract pairs in a state graph (approximately $4n\text{--}6n$ edges for typical tract adjacency graphs), the total computation is:

$$O(n \times (W_1 + W_2)) \approx O(n \times 2\bar{W}) \quad (6)$$

For NC ($n = 2,655$ tracts, approximately 7,000 adjacency edges, $\bar{W} \approx 20$):

$$\text{operations} \approx 7000 \times 40 = 280,000 \quad (7)$$

This is trivially fast; the dominant cost is the initial block-to-tract aggregation and sparse matrix construction. The Phase 2 performance target of under 60 seconds wall time for NC on 8 workers is expected to be achieved comfortably.

3.4 Sparsity Handling

Tracts with no OD data. Some census tracts have no OD records in the LODES file. This occurs for:

- Very low-population tracts (fewer than 3 workers to protect privacy; LODES suppresses records with cell counts below 3).
- Institutional population tracts: prisons, dormitories, military barracks where residents are present but UI-covered commuting is not recorded.
- Tracts that are entirely non-residential (airports, industrial parks, commercial cores with no residential population).

For pairs where one or both tracts have empty distributions, the weighted Jaccard is undefined (denominator is zero). The implementation handles this with a neutral default:

$$\text{sim}(h_1, h_2) = 0.0 \quad \text{if } D(h_1) = \emptyset \text{ or } D(h_2) = \emptyset \quad (8)$$

A similarity of 0.0 corresponds to a neutral edge: the M.4 weight falls to $\alpha \times w_{\text{geo}}(h_1, h_2)$, meaning geographic boundary length governs with the M.4 blend factor applied. This is

the correct behavior: a tract with no commuting data should not be treated as similar or dissimilar to its neighbors; it should simply not participate in the commuting shed signal.

Low-count tracts. Tracts with very few commuting records ($\sum_w D(h)[w] < 10$) have statistically unreliable distributions. The implementation applies a minimum count threshold: tracts with fewer than 10 total workers in the OD file are treated as empty distributions (similarity defaults to 0.0).

3.5 Edge Weight Formula

The M.4 edge weight applies the M-track blend formula:

$$w_{cs}(h_1, h_2) = w_{geo}(h_1, h_2) \times (\alpha + (1 - \alpha) \text{sim}(h_1, h_2)) \quad (9)$$

with default blend factor $\alpha = 0.5$.

The blend preserves geographic boundary length as a baseline constraint and adds commuting shed community signal on top. For tracts with zero commuting similarity (disjoint destination sets), the edge weight is $\alpha \times w_{geo}$, reducing the boundary cost by half. For tracts with perfect commuting similarity (identical destination distributions), the edge weight is w_{geo} , preserving the full geographic cost.

CLI invocation.

```
bisect state --state NC --year 2020 \
    --weights-override commuting-shed \
    --version v_commute_test
```

3.6 Test Invariants

The following L0 unit test invariants must hold for any correct implementation:

same_tract_similarity_is_one. For any home tract h with $\sum_w D(h)[w] > 0$: $\text{sim}(h, h) = 1.0$. (Minimum and maximum are identical, so the ratio is 1.0.)

disjoint_destinations_similarity_is_zero. For two tracts h_1, h_2 where $D(h_1)$ and $D(h_2)$ have no destination tract in common: $\text{sim}(h_1, h_2) = 0.0$.

jaccard_symmetric. For all tract pairs: $\text{sim}(h_1, h_2) = \text{sim}(h_2, h_1)$. (Follows from commutativity of min and max.)

4 Empirical Design

4.1 Experimental Design

Quantitative results for M.4 commuting shed weights are deferred to Phase 2, pending download of LODES OD data for the target states.¹ This section describes the experimental protocol governing Phase 2 execution.

States. Two states are selected to test distinct aspects of the commuting shed signal:

- **Wisconsin** ($k = 8$). WI provides the suburban clustering test. The Milwaukee metro is characterized by distinct commuting sheds: the western suburbs (Waukesha County) commute primarily to Milwaukee and the western suburban employment corridor (Brookfield, Wauwatosa, Menomonee Falls, Pewaukee); the northern suburbs (Ozaukee, Washington counties) commute primarily to Milwaukee and the northern suburban corridor (Glendale, Brown Deer, Menomonee Falls). Both sheds overlap at downtown Milwaukee as a common destination. The prediction: tracts in these suburban sheds co-assign to the same district at $\geq 70\%$ under commuting-shed weights, versus $\leq 50\%$ under geographic weights.
- **North Carolina** ($k = 14$). NC provides the polycentric separation test. The Research Triangle (Wake + Durham + Orange + Chatham counties) and the Charlotte metro (Mecklenburg + Union counties) are two distinct functional economic areas separated by 160 miles of Piedmont geography. Under geographic weights, a plan that bundles some Piedmont tracts with the Triangle or with Charlotte can produce cross-metro district assignments. Under commuting-shed weights, the disjoint destination distributions of Triangle-commuting and Charlotte-commuting tracts should prevent such cross-metro assignments.

All experiments use:

- Algorithm: `standard-bisect` structure
- Seed: 42 (single seed; multi-seed variance designated Phase 3)
- Blend factor: $\alpha = 0.5$
- Year: 2020 decennial census population and 2020 LODES OD

¹Phase 2 will be triggered after LODES OD files are downloaded. The data URL pattern is https://lehd.ces.census.gov/data/lodes/LODES8/{state}/od/{state}_od_main_JT00_2020.csv.gz. Files are downloaded via `bisect fetch -type lodes-od -year 2020`.

- Comparison baseline: `-weights-override geographic` on identical plan configuration

4.2 Primary Metric: Suburban Clustering (Wisconsin)

The WI primary metric is the *Milwaukee suburban co-assignment rate*: the fraction of tract pairs (h_1, h_2) drawn from the Milwaukee suburban counties (Waukesha, Ozaukee, Washington) that are assigned to the same district.

$$\text{SCR} = \frac{\#\{(h_1, h_2) : h_1, h_2 \in \mathcal{S}, \pi(h_1) = \pi(h_2)\}}{\#\{(h_1, h_2) : h_1, h_2 \in \mathcal{S}\}} \quad (10)$$

where $\mathcal{S}$ is the set of Milwaukee suburban tracts and $\pi(h)$ is the district assignment of tract h .

The Phase 2 prediction: $\text{SCR}_{\text{commute}} \geq 0.70$ and $\text{SCR}_{\text{geographic}} \leq 0.50$.[†]

4.3 Primary Metric: Polycentric Separation (North Carolina)

The NC primary metric is the *cross-metro co-assignment rate*: the fraction of tract pairs (h_1, h_2) where h_1 is in the Research Triangle metro and h_2 is in the Charlotte metro that are assigned to the same district.

$$\text{CMR} = \frac{\#\{(h_1, h_2) : h_1 \in \mathcal{T}, h_2 \in \mathcal{C}, \pi(h_1) = \pi(h_2)\}}{\#\{(h_1, h_2) : h_1 \in \mathcal{T}, h_2 \in \mathcal{C}\}} \quad (11)$$

where $\mathcal{T}$ is the set of Research Triangle metro tracts and $\mathcal{C}$ is the set of Charlotte metro tracts.

The Phase 2 prediction: $\text{CMR}_{\text{commute}} < 0.05$ and $\text{CMR}_{\text{geographic}} \leq 0.15$.[†]

A cross-metro co-assignment rate below 5% under commuting-shed weights would confirm that the weighted Jaccard similarity correctly identifies the Triangle and Charlotte as distinct functional economic areas that should not be bundled into the same district.

4.4 Performance Validation

A third metric validates the computational cost claim:

*Wall time for weighted Jaccard computation across all NC tract adjacency edges:
< 60 seconds on 8 workers.*

This is measured by the BISECT internal timer for the weight construction phase, reported in the `bisect state` diagnostic output. The prediction follows from the $O(n \times 2\bar{W})$

complexity analysis in Section 3: with $n \approx 7,000$ NC adjacency edges and $\bar{W} \approx 20$, approximately 280,000 operations are required, a trivially fast computation even on a single thread.[†]

4.5 Phase 2 Timeline

All quantitative results in this section are marked [†]Phase 2 and will be populated when the following prerequisites are met:

1. LODES OD files for NC and WI downloaded via `bisect fetch -type lodes-od -year 2020`.
2. Block-to-tract aggregation completed and sparse commuting distribution matrices stored in Parquet format.
3. L0 test invariants verified for the aggregated distributions (Section 3).
4. Single-seed redistricting runs completed for both states under both `commuting-shed` and `geographic` weight modes.
5. SCR, CMR, and wall time metrics computed from plan output.

5 Legal Usage

5.1 Functional Economic Areas and Communities of Interest

The concept of the functional economic area—a geographic region whose residents share a labor market, a retail trade area, and a common daily economic sphere—has been recognized in redistricting law as a legitimate basis for community of interest claims.

The National Conference of State Legislatures’ 2021 survey of redistricting criteria found that 35 states include communities of interest as a criterion, and that “economic interests” and “shared economic community” are among the most frequently cited community subtypes in commission guidelines and judicial interpretations [National Conference of State Legislatures, 2021]. A commuting shed is a precise quantification of the “shared economic community” concept: the residents of a commuting shed literally share their economic daily life—they compete for the same jobs, drive the same routes, patronize the same establishments, and depend on the same employers.

California commuting communities. California’s Citizens Redistricting Commission criteria cite communities defined by “economic and social interests” [California Secretary of State, 2008]. In its 2011 and 2021 proceedings, the Commission accepted testimony from commuting communities: residents of the Inland Empire testified that their communities shared commuting patterns into the Los Angeles basin, creating economic interdependencies not reflected in county boundaries. Silicon Valley workers testified that the commuting shed from San Jose through Mountain View to Menlo Park defined an economic community distinct from the agricultural Santa Clara Valley south of Morgan Hill. The M.4 algorithm operationalizes these commuting community arguments: tracts with overlapping commuting destinations receive high similarity and heavy edges; tracts commuting to different destinations receive low similarity and light edges.

Colorado trade area communities. Colorado’s redistricting criteria explicitly recognize “trade area” community membership—communities that share a retail trade zone or labor market area [Colorado General Assembly, 2018]. A commuting shed is the labor market equivalent of a trade area: it identifies which residential communities are economically integrated with which employment centers. The Colorado Front Range provides a clear example: communities in the Boulder valley commute to Boulder and Denver; communities in the Fort Collins corridor commute to Fort Collins and the I-25 tech corridor. These are distinct commuting sheds that correspond to distinct communities of interest in redistricting proceedings.

5.2 Airport and Port Communities

Airport and port communities provide some of the clearest examples of commuting-shed community identification in redistricting proceedings.

Airport communities. The area surrounding a major commercial airport is a distinctive commuting community. Residents of tracts near Chicago O’Hare International Airport, Houston George Bush Intercontinental Airport, or Los Angeles International Airport share an employment base (aviation industry, cargo logistics, hotel and hospitality services) and commuting pattern. The LODES OD data captures this concentration: workers in airport-adjacent tracts send flows to the airport employment blocks, creating a tight destination cluster. Under M.4, airport tracts share high weighted Jaccard similarity with each other and low similarity with non-airport tracts commuting to downtown office employment.

This commuting shed community has been recognized in redistricting proceedings. In the 2001 California redistricting, residents near LAX testified before the Special Master that their

community was defined by the airport economy—not by their city of residence (Inglewood, El Segundo, and sections of Los Angeles) or by the county boundary (all three are in Los Angeles County). The court-appointed Special Master’s final map kept the airport industrial and residential community together, in a configuration consistent with M.4’s prediction of high within-shed similarity.

Port communities. Port communities around major shipping facilities have a similarly distinctive commuting character. Residents near the Port of Long Beach/Los Angeles, the Port of Houston-Galveston, and the Port of Baltimore-Newport News all exhibit commuting patterns directed primarily to the port and its logistics hinterland rather than to central business districts. Under M.4, these tracts cluster together based on their shared destinations (port terminals, distribution centers, rail yards) and are appropriately separated from adjacent residential tracts commuting to downtown employment.

The legal recognition of port communities as a distinct economic community of interest has been affirmed in multiple redistricting proceedings. The Maryland redistricting commission in 2011 accepted testimony from port and waterfront industrial community advocates who argued that Baltimore’s waterfront industrial tracts formed a distinct community. The final plan maintained the port community’s cohesion in a way consistent with the M.4 prediction.

5.3 Partisan Neutrality of Commuting Data

LODES OD data contains no electoral, racial, or partisan information. The commute flow records are derived from UI wage records matched to employer geocodes; they record only where workers live and where they work. No information about the political preferences, racial composition, or electoral behavior of the workers is present in the data.

Any partisan effect of commuting-shed weights—such as the polycentric separation of Raleigh and Charlotte metros in the NC prediction—is a consequence of the geographic correlation between commuting patterns and partisan sorting. Urban employment centers that attract commuters from diverse residential areas are a feature of the population landscape, not a product of the algorithm.

Following *Rucho v. Common Cause* [2019], the use of a non-partisan, federally administered dataset to operationalize a legitimate redistricting criterion (communities of interest) satisfies the constitutional standard for non-partisan redistricting even if the resulting plan has partisan effects.

5.4 Expert Witness Deployment Template

Redistricting practitioners deploying M.4 commuting shed weights should structure expert testimony around the following elements:

1. Data provenance declaration. “I used the U.S. Census Bureau’s LODES Origin-Destination Employment Statistics, version 8, for census year 2020 (<https://lehd.ces.census.gov/data/>). This is a federal administrative dataset derived from state Unemployment Insurance wage records matched to residential and workplace geocodes. It records, for each home census block, how many workers residing in that block are employed in each work census block. No voter registration data, election results, racial composition data, or any other socioeconomic information about workers is present in the dataset or used in the computation.”

2. Commuting community identification. “I computed a commuting destination distribution for each census tract: a record of how many workers residing in each tract commute to each employment destination. Two census tracts are in the same commuting shed if they share a substantial fraction of their commuting destinations. This overlap is measured by the weighted Jaccard similarity coefficient: the flow-weighted intersection of commuting destinations divided by the flow-weighted union. District boundaries are drawn preferentially at commuting shed boundaries: tracts with overlapping destinations are kept together; tracts commuting to different economic areas are separated.”

3. Specific community identification. “In [state], the following commuting communities are preserved by the proposed plan: [specific examples, e.g., ‘the Research Triangle Park commuting shed, comprising tracts in Wake, Durham, Orange, and Chatham counties whose workers commute to the RTP technology corridor, is preserved as a cohesive economic community within two contiguous congressional districts. The Charlotte metro commuting shed, comprising Mecklenburg and Union county tracts commuting to the Uptown Charlotte and I-77/I-85 employment corridors, is preserved separately. No district combines Research Triangle and Charlotte commuting shed tracts, reflecting the 160-mile functional economic separation between these areas’].”

4. Criterion alignment. “The [state]’s redistricting criteria require preservation of communities defined by economic and shared-interest considerations. The commuting shed similarity weights directly implement this criterion: communities of workers who share a functional economic area are kept together by the higher edge weights between their residential

tracts.”

6 Conclusion

M.4 operationalizes commuting shed community membership as an edge weight signal through the Census Bureau’s LODES Origin-Destination file. The weighted Jaccard similarity of commuting destination distributions measures the overlap between two tracts’ daily economic spheres: tracts whose workers commute to the same employment centers share a functional economic area and receive heavy edges that METIS avoids cutting.

The signal fills a gap that M.1 (economic character) leaves open. M.1 identifies employment centers—where jobs are concentrated. M.4 identifies which residential communities belong to which employment center’s catchment area. In polycentric metropolitan areas, where multiple large employment centers compete for workers from the same residential zones, M.4 provides the signal that correctly assigns residential tracts to their primary functional economic community.

Relationship to M.1. The complementarity between M.1 and M.4 is a design feature of the M-track series. M.1 fires at residential-employment center boundaries (the RTP campus tract has high JPR, its residential neighbors have low JPR: a strong M.1 cut seam). M.4 fires at commuting shed boundaries (the RTP residential catchment area has high Triangle destination overlap; Charlotte residential tracts have low Triangle destination overlap: a strong M.4 cut seam further from the employment center). Together, M.1 and M.4 define the economic community at two different geographic scales: the employment cluster boundary (M.1) and the residential catchment boundary (M.4). The M.8 composite index [Della-Libera, 2026b] assigns M.1 a weight of 0.20 and M.4 a weight of 0.10, reflecting the judgment that M.4 is a meaningful but more indirect community signal than the direct economic character differentiation of M.1.

Commuting data and the daily life argument. The commuting shed argument for communities of interest is grounded in the observation that daily economic behavior—not formal administrative membership or physical proximity—defines the economic community most people live in. The farmer who drives an hour to a market town, the suburban worker who commutes to an office park, the airport employee who drives the same route to the same terminal every morning: these are the daily economic communities that redistricting commissions are asked to preserve. The LODES OD data makes these communities measurable for the first time in an algorithmic redistricting system.

Phase 2 commitments. Phase 2 will complete the following quantitative results, all currently marked [†]:

1. Milwaukee suburban co-assignment rate (SCR) for WI-8, comparing commuting-shed weights to geographic baseline. Prediction: $\text{SCR}_{\text{commute}} \geq 0.70$ vs. $\text{SCR}_{\text{geographic}} \leq 0.50$.
2. Research Triangle / Charlotte cross-metro co-assignment rate (CMR) for NC-14. Prediction: $\text{CMR}_{\text{commute}} < 0.05$ vs. $\text{CMR}_{\text{geographic}} \leq 0.15$.
3. Weighted Jaccard wall time for NC. Prediction: < 60 seconds on 8 workers.
4. Proportionality gap comparison for WI-8 and NC-14 (secondary metric; predicted to show improvement for NC and null for WI).

Phase 2 depends on LODS OD data download via `bisect fetch -type lods-od -year 2020`. The same LEHD infrastructure used for M.1 WAC data handles this download; the OD file uses the same authentication and the same Parquet caching pipeline.

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